# IGCSE 0607 Practice Questions — E5.7

## Circle theorems II

**Total Questions:** 15 — **Total Marks:** 95 Marks

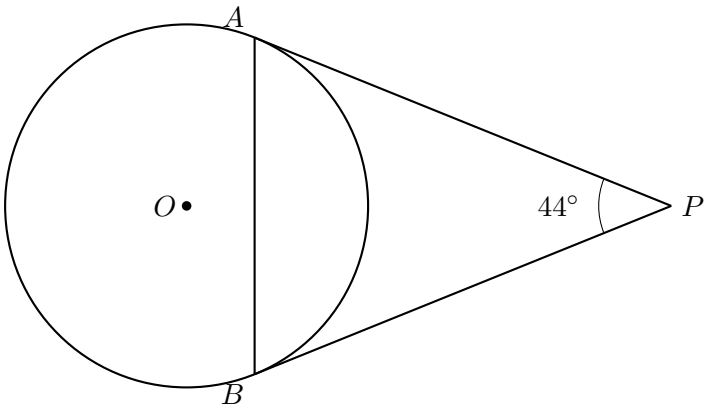
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  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*


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**1** The diagram shows a circle with centre $O$.
$P$ is an external point.
The lines $PA$ and $PB$ are tangents to the circle at the points $A$ and $B$.
Angle $APB = 44°$.




NOT TO SCALE

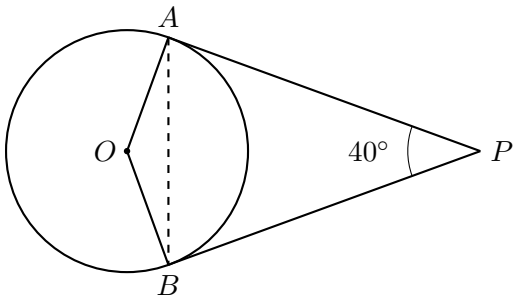
**(a)** Calculate angle $PAB$.

Angle $PAB$ = ..... ° **[2]**

**(b)** Calculate angle $AOB$.

Angle $AOB$ = ..... ° **[2]**

**2** $P$ is a point outside a circle, centre $O$.
$PA$ and $PB$ are tangents to the circle at $A$ and $B$.
Angle $APB = 40°$.




**(a)** Write down the geometrical reason why triangle $PAB$ is isosceles.

..... **[1]**

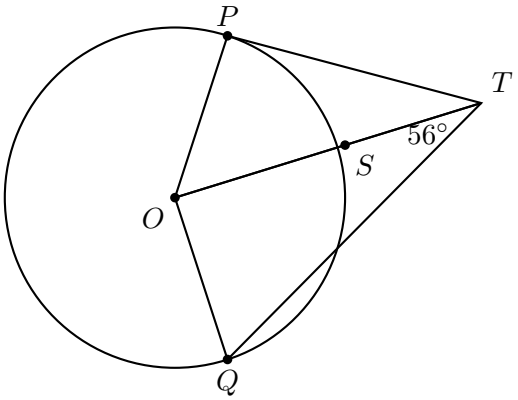
**(b)** Calculate angle $PAB$.

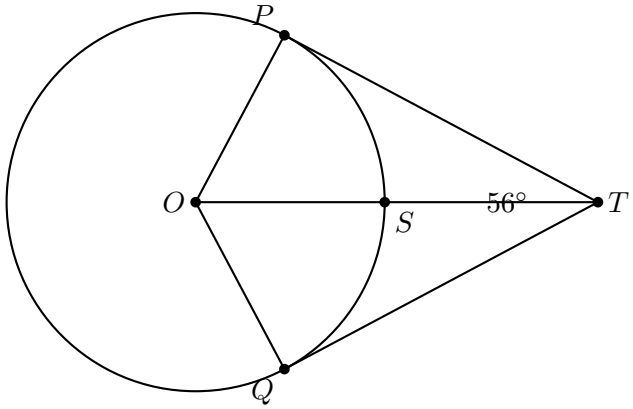
Angle $PAB =$ ..... ° **[1]**

**(c)** Calculate angle $AOB$.

Angle $AOB =$ ..... ° **[2]**

**3** $P$ and $Q$ are points on a circle, centre $O$.
$TP$ and $TQ$ are tangents to the circle from an external point $T$.
The line $TO$ intersects the circle at $S$.
Angle $PTQ = 56°$.




**(a)** Calculate angle $POT$.

Angle $POT =$ ..... ° **[2]**

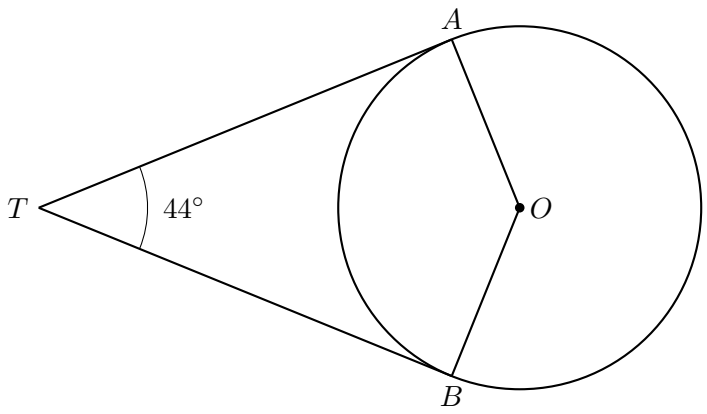
**(b)** Explain why triangle $TPQ$ is isosceles.

..... **[1]**

**(c)** Calculate angle $TPQ$.

Angle $TPQ =$ ..... ° **[1]**

**4** The diagram shows a circle, centre $O$.
$TA$ and $TB$ are tangents to the circle at $A$ and $B$.
Angle $ATB = 44°$.




**(a)** Work out angle $AOB$.

Angle $AOB =$ ..... ° **[2]**

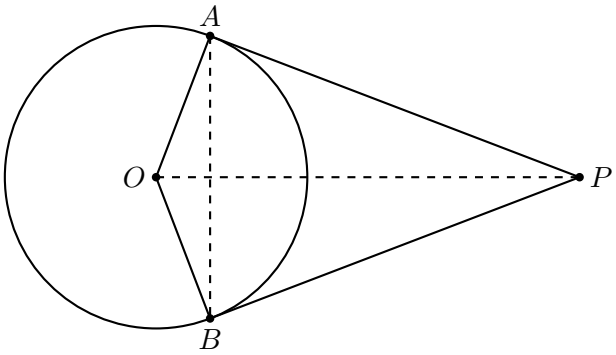
**(b)** Work out angle $OAB$.

Angle $OAB =$ ..... ° **[2]**

**(c)** Write down the geometrical reason why the length of $TA$ is equal to the length of $TB$.

..... **[1]**

**5** $P$ is an external point.
$PA$ and $PB$ are tangents to a circle, centre $O$.
The line $OP$ intersects the chord $AB$ at right angles.
$OA = 7$ cm and $PA = 24$ cm.




**(a)** Write down the length of $PB$.

..... cm **[1]**

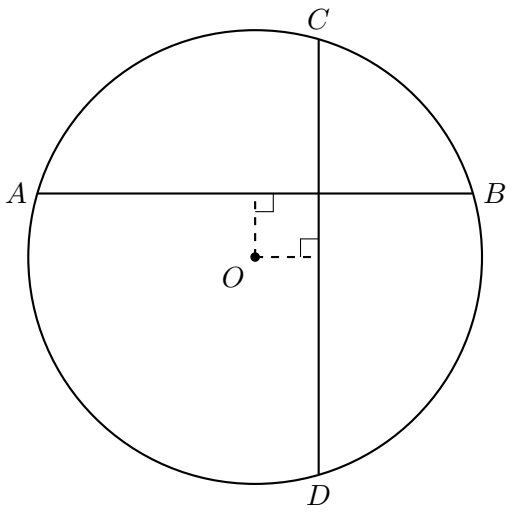
**(b)** Calculate the length of $OP$.

..... cm **[2]**

**(c)** Calculate the shortest distance from $O$ to the chord $AB$.

..... cm **[3]**

**6** The diagram shows a circle, centre $O$, with radius 25 cm.
$AB$ and $CD$ are two chords of the circle such that $AB = CD$.
The perpendicular distance from $O$ to $AB$ is $(x + 3)$ cm.
The perpendicular distance from $O$ to $CD$ is $(3x - 5)$ cm.



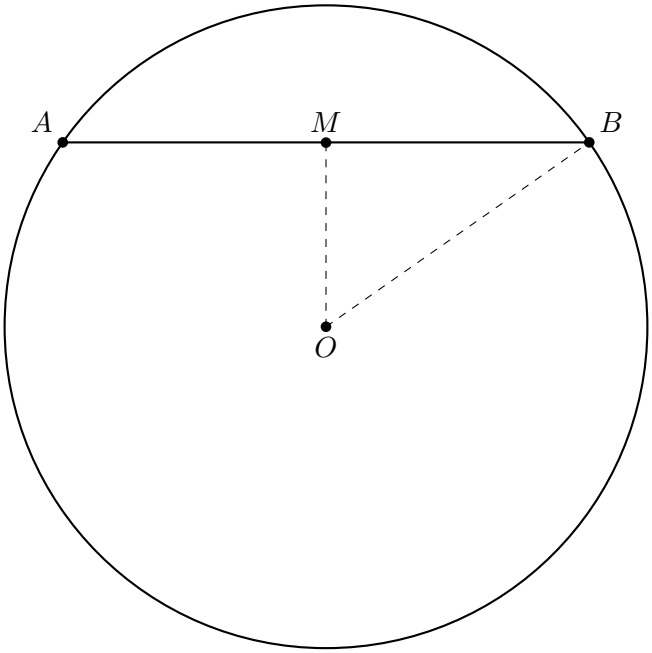

**(a)** Find the value of $x$.

$x =$ ..... **[2]**

**(b)** Calculate the length of the chord $AB$.

$AB =$ ..... cm **[3]**

**7** A circle, centre $O$, has a radius of 8.5 cm.
A chord $AB$ has length 14 cm.
$M$ is the midpoint of $AB$.



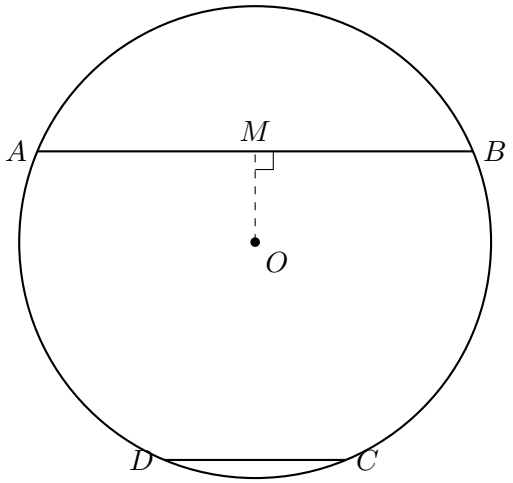

**(a)** Calculate the distance $OM$.

$OM =$ ..... cm **[3]**

**(b)** Another chord $CD$ is drawn such that it is parallel to $AB$ and is at a distance of 3 cm from $O$.
Calculate the length of the chord $CD$.

$CD =$ ..... cm **[2]**

**8** The diagram shows a circle, centre $O$, with radius 13 cm.
$AB$ is a chord of the circle of length 24 cm.
The point $M$ is the midpoint of the chord $AB$.
Another chord $CD$ is drawn inside the circle.




NOT TO SCALE

**(a)** Calculate the perpendicular distance, $OM$, from the centre of the circle to the chord $AB$.

$OM =$ ..... cm **[3]**

**(b)** The perpendicular distance from the centre $O$ to the chord $CD$ is also 5 cm.
Write down the length of the chord $CD$.

$CD =$ ..... cm **[1]**

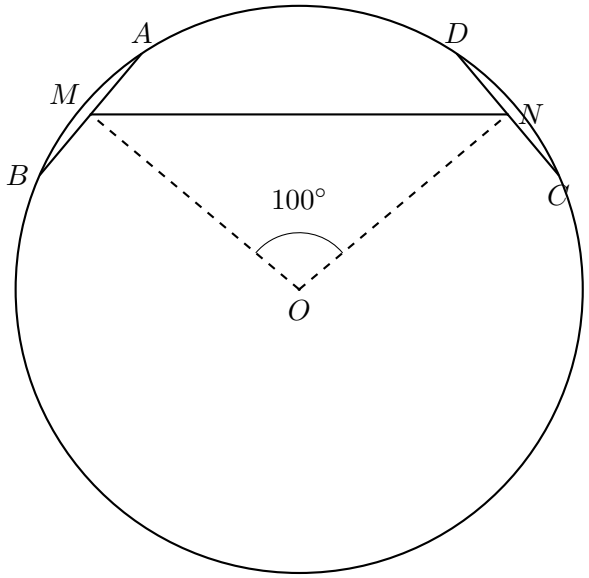
**(c)** Calculate the area of triangle $OAB$.

Area =

.....

$cm^2$ **[2]**

**9** $A$, $B$, $C$ and $D$ are points on a circle, centre $O$ and radius 25 cm.
The chords $AB$ and $CD$ each have a length of 14 cm.
$M$ is the midpoint of $AB$ and $N$ is the midpoint of $CD$.
Angle $MON = 100°$.




**(a)** Show that $OM = 24$ cm.

**[2]**

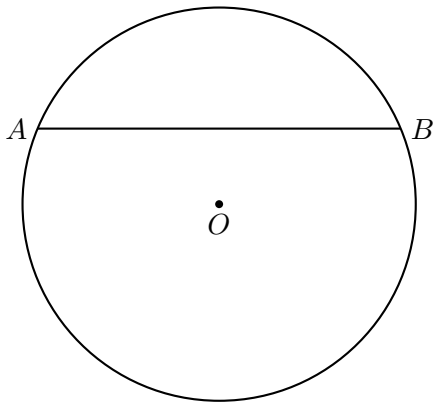
**(b)** Calculate the length of $MN$.

..... cm **[3]**

**(c)** Calculate the area of triangle $MON$.

..... $cm^2$ **[2]**

**10** The diagram shows a circle, centre $O$, with radius 13 cm.
$AB$ is a chord of the circle of length 24 cm.




**(a)** Calculate the shortest distance from $O$ to the chord $AB$.

..... *cm* **[3]**

**(b)** $CD$ is another chord of the circle.
$CD$ is parallel to $AB$ and has length 24 cm.
Write down the shortest distance from $O$ to $CD$.

..... *cm* **[1]**

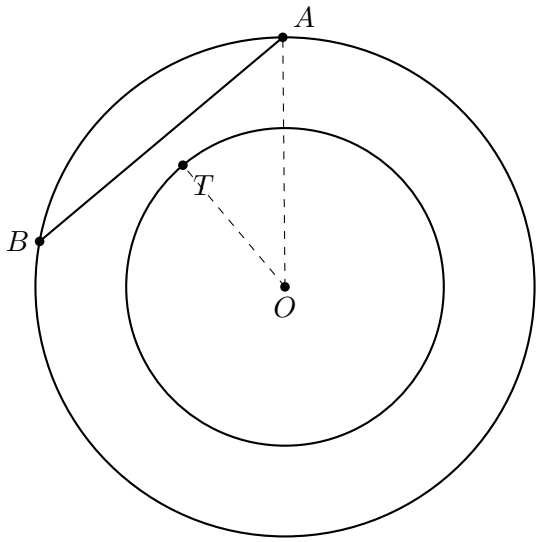
**(c)** $EF$ is a third chord of the circle.
The shortest distance from $O$ to $EF$ is 12 cm.
Calculate the length of the chord $EF$.

..... *cm* **[3]**

**(d)** State the geometrical property of a circle used to find the length of half the chord in part (a).

..... **[1]**

**11** Two circles, $C_1$ and $C_2$, have a common centre $O$.
The radius of $C_1$ is $r$ cm and the radius of $C_2$ is $R$ cm, where $R > r$.
A chord $AB$ of the larger circle $C_2$ is tangent to the smaller circle $C_1$ at the point $T$.
The length of the chord $AB$ is 24 cm and the area of the annular region between the two circles is $k\pi$ cm$^2$.




**(a)** Show that $R^2 - r^2 = 144$.

**[3]**

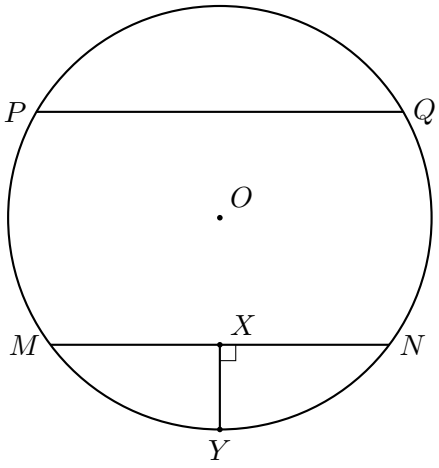
**(b)** Find the value of $k$.

$k =$ ..... **[2]**

**(c)** A second chord $CD$ of the larger circle $C_2$ has length 24 cm.
State, with a reason, the distance from $O$ to the chord $CD$.

..... **[2]**

**12** A circle has centre $O$ and radius 10 cm.
$MN$ is a chord of length 16 cm.
The perpendicular bisector of $MN$ intersects $MN$ at $X$ and the minor arc $MN$ at $Y$.
$PQ$ is a second chord which is parallel to $MN$.
The perpendicular distance between $MN$ and $PQ$ is 11 cm.




**(a)** Explain why the line containing $XY$ passes through the centre $O$.

..... **[1]**

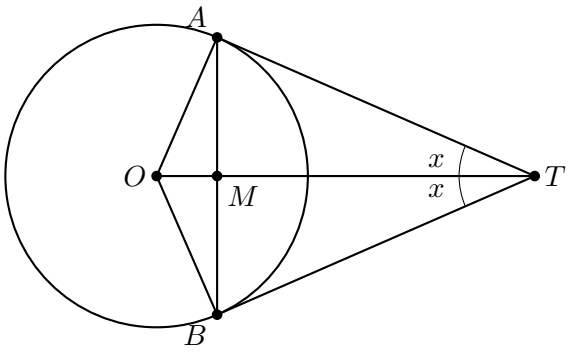
**(b)** Calculate the length of $XY$.

$XY =$ ..... cm **[3]**

**(c)** Calculate the length of the chord $PQ$.

$PQ =$ ..... cm **[4]**

**13** $TA$ and $TB$ are tangents to a circle, centre $O$ and radius $r$, at the points $A$ and $B$.
The line $TO$ intersects the chord $AB$ at $M$.
Angle $ATO$ = Angle $BTO = x$.




**(a)** Show that the length of the chord $AB$ is $2r\cos x$.

**[2]**

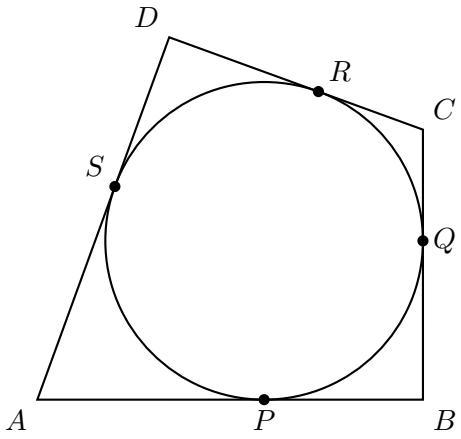
**(b)** A second chord $BC$ is drawn such that $BC = AB$.
Write down the shortest distance from $O$ to the chord $BC$, in terms of $r$ and $x$.

..... **[1]**

**(c)** The area of triangle $OAB$ is equal to the area of triangle $ATB$.
Find the exact value of $x$.

$x =$ ..... $^\circ$ **[5]**

**14** A circle is inscribed in a quadrilateral $ABCD$.
The circle touches the sides $AB$, $BC$, $CD$, and $DA$ at the points $P$, $Q$, $R$, and $S$ respectively.




**(a)** Write down the geometrical reason why $AP = AS$.

..... **[1]**

**(b)** Show that $AB + CD = AD + BC$.

**[3]**

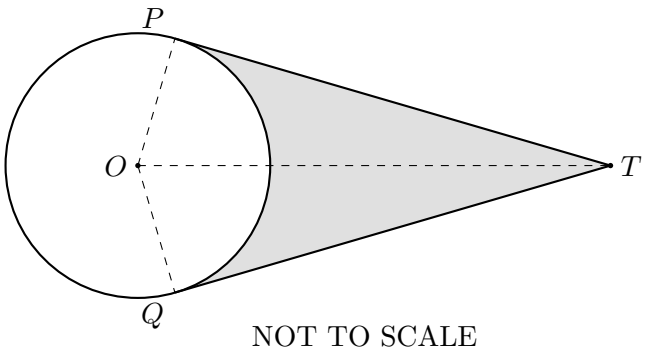
**(c)** Given that $AB = (2x + 5)$ cm, $BC = (3x - 2)$ cm, $CD = (x + 8)$ cm and $AD = (2x + 1)$ cm, work out the value of $x$.

$x =$ ..... **[3]**

**(d)** Calculate the perimeter of the quadrilateral $ABCD$.

Perimeter = ..... $cm$ **[2]**

**15** The diagram shows a circle with centre $O$ and radius 7 cm.
The lines $TP$ and $TQ$ are tangents to the circle at the points $P$ and $Q$.
The distance from the centre $O$ to the external point $T$ is 25 cm.




NOT TO SCALE

**(a)** Show that the length of the tangent $TP$ is 24 cm.

**[2]**

**(b)** Calculate angle $PTQ$.

Angle $PTQ$ = ..... ° **[3]**

**(c)** Calculate the area of the shaded region bounded by the tangents $TP$, $TQ$ and the minor arc $PQ$.

Area =

.....

$cm^2$ **[4]**